



CS 340 README Template

About the Project

Grazioso Salvare is an innovative international rescue-animal training company that identifies dogs suitable for search-and-rescue training. This software application can work with existing data from animal shelters to identify and categorize available dogs for training. The application includes a database.

Motivation

Grazioso Salvare needs to identify dogs with specific profiles suitable for different types of rescue missions, such as water rescue, mountain rescue, disaster recovery, and scent tracking. The project aims to improve this process by using data from local animal shelters.

Getting Started

Installation prerequisites:

- **MongoDB:** We need MongoDB to store and manage our data in a document-oriented database. You can download it from the [official MongoDB website](#).
- **Python 3+:** Programming language for writing and executing our data processing scripts. You can download it from the [official Python website](#).
- **Jupyter Notebook:** Use Jupyter Notebook to write and run our Python code, visualize data, and much more. You can install it using pip (after installing Python) by following the steps in the [official Jupyter website](#).

Follow the Installation steps to setup the project.

Installation

1. Clone the repository or download the files. This repository includes the following files:
 - animal_shelter.py
 - ProjectTwoDashboard.ipynb
 - Grazioso Salvare Logo.png
2. Install **pymongo** to connect our Python code to the MongoDB database and perform CRUD database operations from our scripts:

```
pip install pymongo
```

3. Import the **aac_shelter_outcomes.csv** file into MongoDB:

```
mongoimport --username="${MONGO_USER}" --password="${MONGO_PASS}" --port=${MONGO_PORT} --host=${MONGO_HOST} --db AAC --collection animals --type csv --headerline --authenticationDatabase admin --file path/to/aac_shelter_outcomes.csv
```

(Note: If the command throws an error, make sure that the .csv file directory is correct and that you are using the correct MongoDB user).

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).

```

jeanfarfan_snhu@nv-snhu7-l02: /usr/local/datasets
(base) jeanfarfan_snhu@nv-snhu7-l02:/usr/local/datasets$ mongoimport --username=
"${MONGO_USER}" --password="${MONGO_PASS}" --port=${MONGO_PORT} --host=${MONGO_H
OST} --db AAC --collection animals --type csv --headerline --authenticationDatab
ase admin --drop ./aac_shelter_outcomes.csv
2024-05-30T21:36:36.134+0000    connected to: mongodb://nv-desktop-services.appo
rto.com:32308/
2024-05-30T21:36:36.135+0000    dropping: AAC.animals
2024-05-30T21:36:37.281+0000    10000 document(s) imported successfully. 0 docum
ent(s) failed to import.
(base) jeanfarfan_snhu@nv-snhu7-l02:/usr/local/datasets$

```

4. Edit the connection variables in *animal_shelter.py* file to match your MongoDB:

```

USER = 'your_user'
PASS = 'your_password'
HOST = 'your_host'
PORT = your_port
DB = 'AAC'
COL = 'animals'

```

Create a Mongo user with read and write privileges for our database (optional)

1. Open the terminal and enter the MongoDB shell with the following command:

```
mongosh
```

2. Use admin database:

```
use admin
```

3. Create user:

```
db.createUser({user: "your_username", pwd: "your_password", roles: [{role: "readWrite", db: "AAC"}]})
```

```

jeanfarfan_snhu@nv-snhu7-l01: ~
(base) jeanfarfan_snhu@nv-snhu7-l01:~$ mongosh
Current Mongosh Log ID: 66529d807f915f07797428de
Connecting to:      mongodb://<credentials>@nv-desktop-services.apporto.com:
32308/?directConnection=true&appName=mongosh+1.8.0
Using MongoDB:      6.0.13
Using Mongosh:      1.8.0

For mongosh info see: https://docs.mongodb.com/mongosh-shell/

-----
The server generated these startup warnings when booting
2024-05-26T02:10:13.307+00:00: Using the XFS filesystem is strongly recommend
ed with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodno
tes-filesystem
2024-05-26T02:10:15.393+00:00: Failed to read /sys/kernel/mm/transparent_huge
page/defrag
2024-05-26T02:10:15.393+00:00: vm.max_map_count is too low
-----

test> db.runCommand({connectionStatus:1})
{
  authInfo: {
    authenticatedUsers: [ { user: 'root', db: 'admin' } ],
    authenticatedUserRoles: [ { role: 'root', db: 'admin' } ]
  },
  ok: 1
}
test> use admin
switched to db admin
admin> db.createUser({user: "aacuser", pwd: "Ελληνικα", roles: [{role: "readWrit
e", db: "AAC"}]})
{ ok: 1 }
admin> exit
(base) jeanfarfan_snhu@nv-snhu7-l01:~$

```

4. Exit MongoDB shell and change Mongo user in the terminal:

MONGO_USER=your_username

MONGO_PASS=your_password

5. Try to open the MongoDB shell and execute the following command for verification:

```
db.runCommand({connectionStatus:1})
```

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).

```

(base) jeanfarfan_snhu@nv-snhu7-l01: ~
(base) jeanfarfan_snhu@nv-snhu7-l01:~$ MONGO_USER=aacuser
(base) jeanfarfan_snhu@nv-snhu7-l01:~$ MONGO_PASS=Eλληνικα
(base) jeanfarfan_snhu@nv-snhu7-l01:~$ printenv | grep -i mongo
MONGO_USER=aacuser
MONGO_HOST=nv-desktop-services.apporto.com
MONGO_PASS=Eλληνικα
MONGO_PORT=32308
(base) jeanfarfan_snhu@nv-snhu7-l01:~$ mongosh
Current Mongosh Log ID: 6652a2d2db468e81bc78cdb1
Connecting to:      mongodb://<credentials>@nv-desktop-services.apporto.com:
32308/?directConnection=true&appName=mongosh+1.8.0
Using MongoDB:      6.0.13
Using Mongosh:      1.8.0

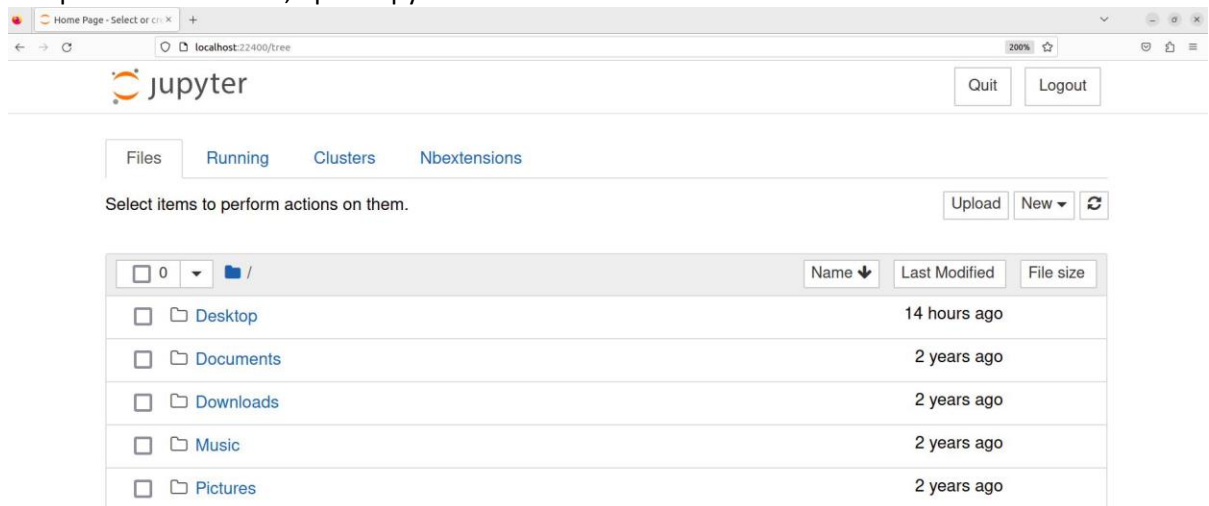
For mongosh info see: https://docs.mongodb.com/mongodb-shell/

test> db.runCommand({connectionStatus:1})
{
  authInfo: {
    authenticatedUsers: [ { user: 'aacuser', db: 'admin' } ],
    authenticatedUserRoles: [ { role: 'readWrite', db: 'AAC' } ]
  },
  ok: 1
}

```

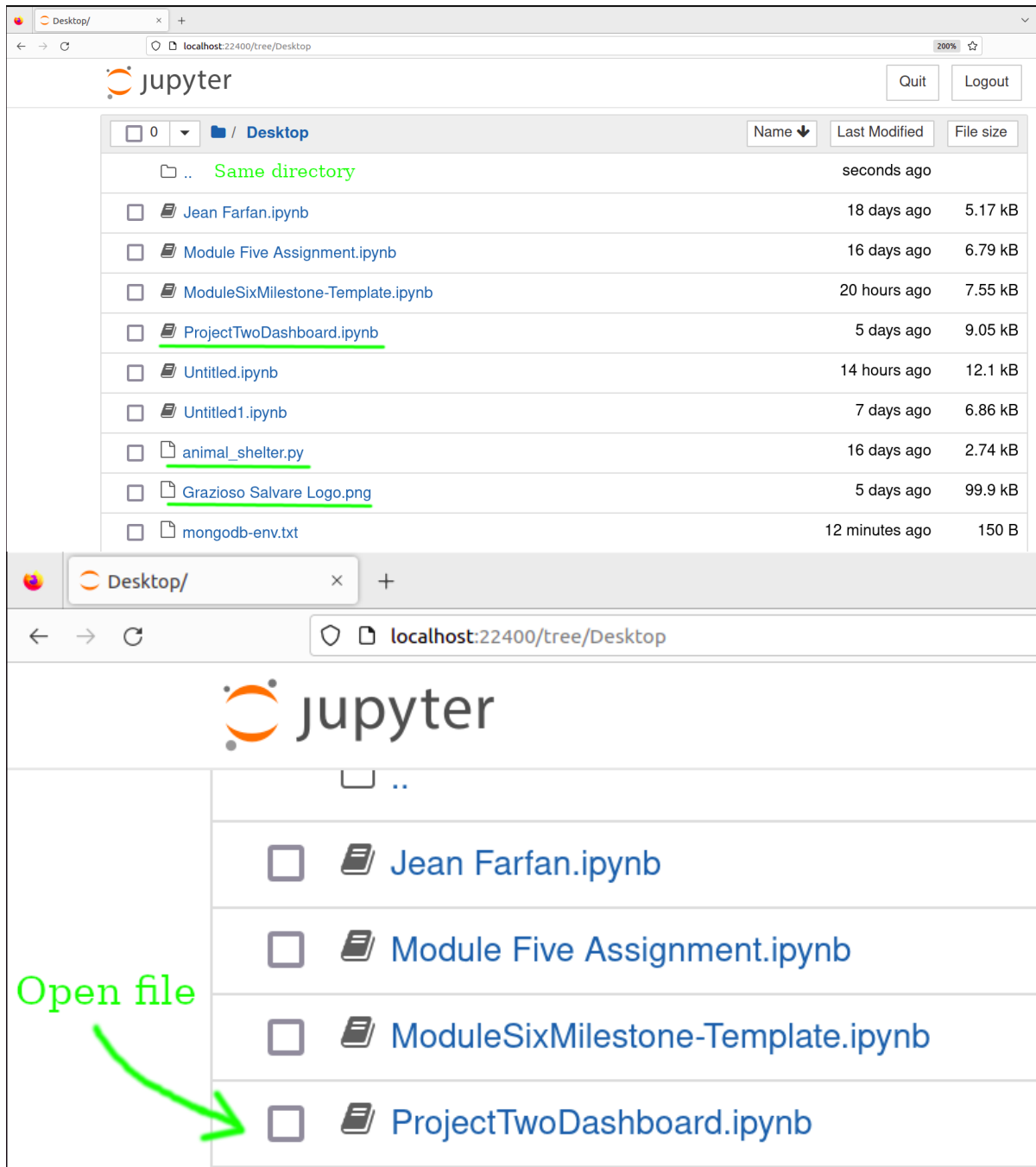
Open dashboard

1. To open the dashboard, open Jupyter Notebook.



2. Search and open the **ProjectTwoDashboard.ipynb** file. Make sure that all the files (**ProjectTwoDashboard.ipynb**, **Grazioso Salvare Logo.png**, and **animal_shelter.py**) are in the same directory.

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).



The screenshot shows the JupyterLab interface with the file browser pane open to the 'Desktop' directory. The file list includes:

Name	Last Modified	File size
..	Same directory	seconds ago
Jean Farfan.ipynb	18 days ago	5.17 kB
Module Five Assignment.ipynb	16 days ago	6.79 kB
ModuleSixMilestone-Template.ipynb	20 hours ago	7.55 kB
<u>ProjectTwoDashboard.ipynb</u>	5 days ago	9.05 kB
Untitled.ipynb	14 hours ago	12.1 kB
Untitled1.ipynb	7 days ago	6.86 kB
animal_shelter.py	16 days ago	2.74 kB
<u>Grazioso Salvare Logo.png</u>	5 days ago	99.9 kB
mongodb-env.txt	12 minutes ago	150 B

A green arrow points to the 'ProjectTwoDashboard.ipynb' file with the text 'Open file'.

- Run the file to show the dashboard. This will launch the dashboard on a local server at <http://127.0.0.1:8050/>.

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).

jupyter ProjectTwoDashboard Last Checkpoint: Last Wednesday at 7:05 PM (unsaved changes)

File Edit View Insert Cell Kernel Widgets Help

Run

Run project

```
In [1]: # Setup the Jupyter version of Dash
from jupyter_dash import JupyterDash

# Configure the necessary Python module imports for dashboard components
import dash_leaflet as dl
from dash import dcc
from dash import html
import plotly.express as px
from dash import dash_table
from dash.dependencies import Input, Output, State
import base64

# Configure OS routines
import os

# Configure the plotting routines
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# CRUD Python module
from animal_shelter import AnimalShelter
```

ProjectTwoDashboard Last Checkpoint: Last Wednesday

View Insert Cell Kernel Widgets Help

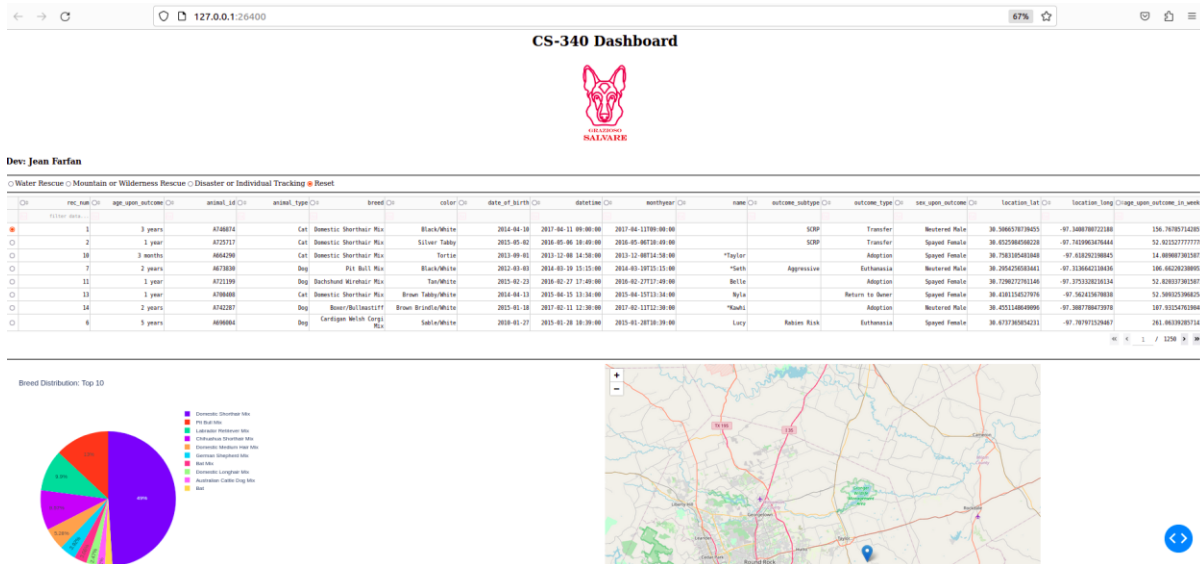
Run

```
children=[
    dl.Tooltip(
    dl.Popup([
        html.H1
        html.P(
    ])
    ])
]

app.run_server(debug=True)
```

Dash app running on <http://127.0.0.1:26400/>

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).



Usage

The dashboard developed for Grazioso Salvare uses data from local animal shelters to help identify dogs suitable for various rescue missions. This dashboard provides an interactive interface to filter and visualize the data. This dashboard allows users to filter dogs based on the type of rescue mission. Also, it displays the filtered data in an interactive data table and a pie chart for better analysis and decision-making. Finally, it supports CRUD (Create, Read, Update, and Delete) operations on the dog data stored in MongoDB.

Use interactive radio items to filter the data based on the type of rescue mission. The data table and charts will update dynamically to reflect the filtered data.

To perform Create, Read, Update, and Delete operations on the data, you need to write code in **ProjectTwoDashboard.ipynb** or in a new Python file. Ensure the MongoDB connection variables in `animal_shelter.py` are correctly configured to match your MongoDB setup. Below is an example of how to use CRUD methods using the provided **animal_shelter.py**.

Code Example

```
from animal_shelter import AnimalShelter

shelter = AnimalShelter() # Create an instance of AnimalShelter

# Example animal document
animal = {
    "age_upon_outcome": '99 years',
    "animal_id": 'B666777',
    "animal_type": 'Cuy',
    "breed": 'Domestic Shorthair Mix',
    "color": 'Orange',
```

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).



```
"date_of_birth": '1925-06-13',
"datetime": '2018-04-11 09:00:00',
"monthyear": '2019-04-11T09:00:00',
"name": 'Gamo',
"outcome_subtype": 'SCRP',
"outcome_type": 'Transfer',
"sex_upon_outcome": 'Neutered Male',
"location_lat": 30.5066578739454,
"location_long": -97.3408780722187,
"age_upon_outcome_in_weeks": 156.767857142856
}
```

```
# Create
# Create a new animal
print(shelter.create(animal))

# Find
# Read an animal
results = shelter.read({"animal_id": "B666777"})
for k in results:
    print(k)

# Update
# First, key/value to find
criteria = {"animal_id" : "B666777"}
# Then, key/value to modify
updateData = {"age_upon_outcome": '77 years'}
# print modified count
print(shelter.update(criteria, updateData))
# Find the updated animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
    print (k)
# Delete
# Delete animal with animal_id B666777
print(shelter.delete({"animal_id" : "B666777"}))
# Find deleted animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
    print (k)
```

Tests

To validate the functionality, launch the dashboard and use the interactive widgets to filter data, and perform CRUD operations via the provided interface. This will ensure that the application meets the requirements for identifying and categorizing dogs suitable for various rescue missions.

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).



Screenshots

```
In [1]: from animal_shelter import AnimalShelter
```

```
In [2]: shelter = AnimalShelter() # class object
animal = { # document
    "age_upon_outcome": '99 years',
    "animal_id": 'B666777',
    "animal_type": 'Cuy',
    "breed": 'Domestic Shorthair Mix',
    "color": 'Orange',
    "date_of_birth": '1925-06-13',
    "datetime": '2018-04-11 09:00:00',
    "monthyear": '2019-04-11T09:00:00',
    "name": 'Gamo',
    "outcome_subtype": 'SCRIP',
    "outcome_type": 'Transfer',
    "sex_upon_outcome": 'Neutered Male',
    "location_lat": 30.5066578739454,
    "location_long": -97.3408780722187,
    "age_upon_outcome_in_weeks": 156.767857142856
}
```

```
In [3]: print(shelter.create(animal))
```

True

```
In [4]: results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
    print(k)

{'_id': ObjectId('6659568f6e1cd69e37204665'), 'age_upon_outcome': '99 years', 'animal_id': 'B666777', 'animal_type': 'Cuy', 'breed': 'Domestic Shorthair Mix', 'color': 'Orange', 'date_of_birth': '1925-06-13', 'datetime': '2018-04-11 09:00:00', 'monthyear': '2019-04-11T09:00:00', 'name': 'Gamo', 'outcome_subtype': 'SCRIP', 'outcome_type': 'Transfer', 'sex_upon_outcome': 'Neutered Male', 'location_lat': 30.5066578739454, 'location_long': -97.3408780722187, 'age_upon_outcome_in_weeks': 156.767857142856}
```

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).

```
In [13]: # First, key/value to find
criteria = {"animal_id" : "B666777"}
# Then, key/value to modify
updateData = {"age_upon_outcome": '77 years'}

In [14]: print(shelter.update(criteria, updateData))
1

In [15]: # Find the updated animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
    print (k)

{'_id': ObjectId('66625b88ccd5db01393ede82'), 'age_upon_outcome': '77 years', 'animal_id': 'B666777', 'animal_type': 'Cuy', 'breed': 'Domestic Shorthair Mix', 'color': 'Orange', 'date_of_birth': '1925-06-13', 'datetime': '2018-04-11 09:00:00', 'monthyear': '2019-04-11T09:00:00', 'name': 'Gamo', 'outcome_subtype': 'SCRIP', 'outcome_type': 'Transfer', 'sex_upon_outcome': 'Neutered Male', 'location_lat': 30.5066578739454, 'location_long': -97.3408780722187, 'age_upon_outcome_in_weeks': 156.767857142856}

In [16]: # Delete animal with animal_id B666777
print(shelter.delete({"animal_id" : "B666777"}))
1

In [17]: # Find deleted animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
    print (k)

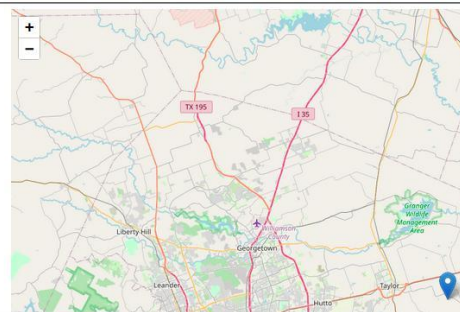
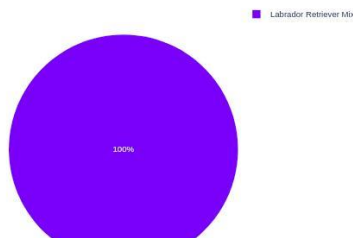
In [ ]:
```

Dev: Jean Farfan

Water Rescue Mountain or Wilderness Rescue Disaster or Individual Tracking Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
filter data...													
36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow		2015-07-06 11:33:00	2015-07-06T11:33:00			Medical	Euthanasia	Intact Female
732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White		2017-07-25 14:59:00	2017-07-25T14:59:00		*Catalina	Return to Owner	Intact Female	
1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black		2016-08-30 14:12:00	2016-08-30T14:12:00		Pirata	Return to Owner	Intact Female	
1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White		2016-12-23 17:13:00	2016-12-23T17:13:00		Mika	Adoption	Intact Female	
1757	7 months	A742767	Dog	Labrador Retriever Mix	Black		2017-02-14 15:20:00	2017-02-14T15:20:00		Marley	Return to Owner	Intact Female	
1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White		2017-12-03 13:09:00	2017-12-03T13:09:00			Partner	Transfer	Intact Female
2041	2 years	A702745	Dog	Labrador Retriever Mix	Black		2015-05-22 11:45:00	2015-05-22T11:45:00		Abigail	Return to Owner	Intact Female	
2225	2 years	A757341	Dog	Labrador Retriever Mix	Black/White		2017-10-03 12:27:00	2017-10-03T12:27:00		19	Partner	Transfer	Intact Female

Breed Distribution: Top 10



Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).

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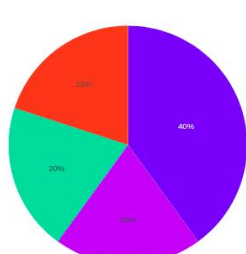
GIBAZIONE
SALVARE

Dev: Jean Farfan

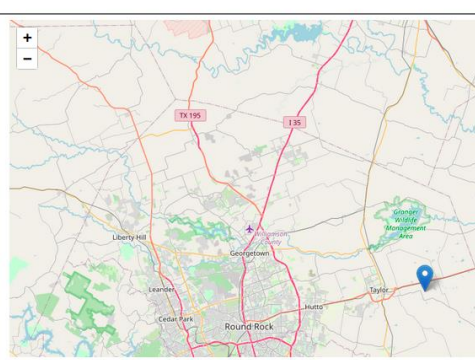
🔹 Water Rescue 🔸 Mountain or Wilderness Rescue 🔸 Disaster or Individual Tracking 🔸 Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
🔸	3130	2 years	A721834	Dog	Siberian Husky	Brown/White	2014-03-05	2016-03-23 16:23:00	2016-03-23T16:23:00		Suffering	Euthanasia	Intact Male 30.
🔹	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24:00	2015-08-02T17:24:00	Papa		Return to Owner	Intact Male 30.
🔹	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00	2017-09-23T11:23:00	Zeke		Return to Owner	Intact Male 30.
🔹	6191	2 years	A704181	Dog	Siberian Husky	Black/White	2013-06-01	2015-06-02 16:41:00	2015-06-02T16:41:00	Lobo		Return to Owner	Intact Male 30.
🔹	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male 3.

Breed Distribution: Top 10



- Siberian Husky
- Alaskan Malamute
- Rottweiler
- German Shepherd



← → ↻
📍 127.0.0.1:26400

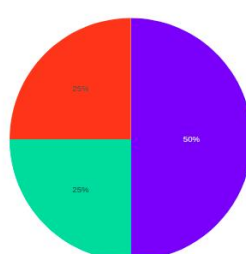
GIBAZIONE
SALVARE

Dev: Jean Farfan

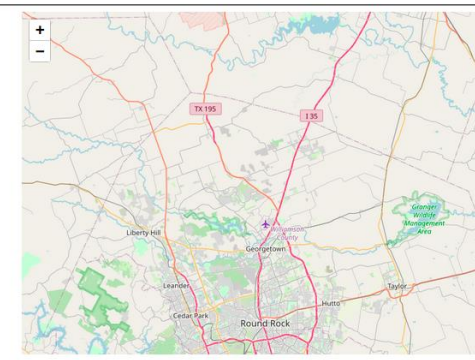
🔹 Water Rescue 🔹 Mountain or Wilderness Rescue 🔸 Disaster or Individual Tracking 🔸 Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
🔸	2987	4 years	A694614	Dog	Rottweiler	Black/Brown	2011-01-01	2015-01-01 14:25:00	2015-01-01T14:25:00	Striker		Return to Owner	Intact Male
🔹	3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43:00	2015-09-22T15:43:00	Boomer		Return to Owner	Intact Male
🔹	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00	2017-09-23T11:23:00	Zeke		Return to Owner	Intact Male
🔹	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male

Breed Distribution: Top 10



- Rottweiler
- Bloodhound
- German Shepherd



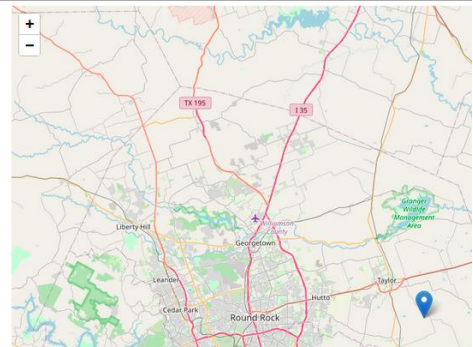
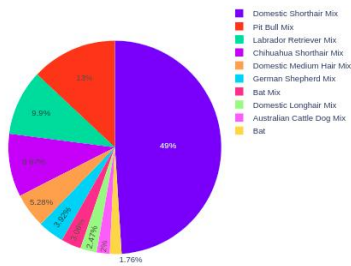
Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).

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○ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster or Individual Tracking ● Reset

filter data...	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
●	1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-16	2017-04-11 09:09:00	2017-04-11T09:00:00		SCRP	Transfer	Neutered Male
○	2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00		SCRP	Transfer	Spayed Female
○	10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adoption	Spayed Female
○	7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19T15:15:00	*Seth	Aggressive	Euthanasia	Neutered Male
○	11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spayed Female
○	13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown Tabby/White	2014-04-13	2015-04-15 13:34:00	2015-04-15T13:34:00	Nyla		Return to Owner	Spayed Female
○	14	2 years	A742287	Dog	Boxer/Bulldog Mix	Brown Brindle/White	2015-01-18	2017-02-11 12:30:00	2017-02-11T12:30:00	*Kawhi		Adoption	Neutered Male
○	6	5 years	A696084	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-28 10:39:00	2015-01-28T10:39:00	Lucy	Rabies Risk	Euthanasia	Spayed Female

Breed Distribution: Top 10



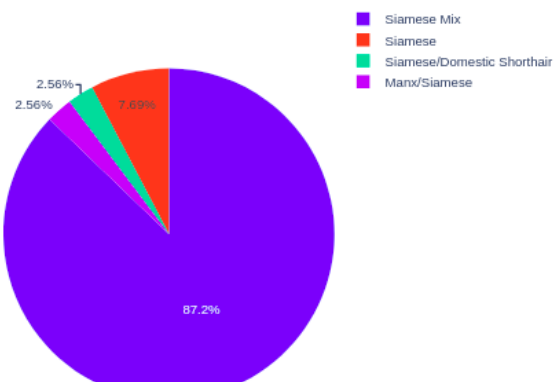


☐ Water Rescue ☐ Mountain or Wilderness Rescue ☐ Disaster or Individual Tracking ☒ Reset

☐	rec_num	☐	age_upon_outcome	☐	animal_id	☐	animal_type	☐	breed	☐	color	☐	date_of_birth
☐	filter data...	☐		☐	A6	☐	Cat	☐	Siamese	☐		☐	
☒	616		5 months		A684362		Cat		Siamese Mix		Tortie Point		2014-05-26
☐	843		1 month		A680409		Cat		Siamese Mix		Flame Point		2014-05-02
☐	1074		1 year		A667027		Cat		Siamese Mix		Flame Point		2012-11-11
☐	1350		2 months		A683412		Cat		Siamese Mix		Flame Point		2014-05-10
☐	1472		3 years		A665943		Cat		Siamese Mix		Lynx Point		2010-10-24
☐	1492		17 years		A698600		Cat		Siamese Mix		Lynx Point		1998-03-14
☐	1623		2 years		A696517		Cat		Siamese Mix		Lynx Point		2013-02-06
☐	1706		10 years		A672606		Cat		Siamese Mix		Lynx Point		2004-02-16



Breed Distribution: Top 10



Contact

Your name: Jean Farfan

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).